

BLUESTOCK FINTECH

DATA ANALYST INTERNSHIP

MUTUAL FUND ANALYTICS PLATFORM

A Complete Data Engineering & Business Intelligence Capstone

Submitted by

SHIVANAND METRI

PGDM — Finance & Business Analytics

M.S. Ramaiah Institute of Management, Bengaluru

Internship Period: June 2026

Supervising Organisation: Bluestock Fintech .in

EXECUTIVE SUMMARY

This report documents the complete design, implementation, and analytical findings of a Mutual Fund Analytics Platform developed during a five-day Data Analyst Internship at Bluestock Fintech Pvt. Ltd. The platform ingests, processes, and analyses data from ten publicly available datasets representing India's Rs. 81 lakh crore mutual fund industry.

The project was executed across five structured phases: (1) data ingestion and ETL pipeline development; (2) PostgreSQL database design and loading; (3) exploratory data analysis; (4) performance and risk analytics; and (5) Power BI dashboard development. All five phases were completed within the five-day internship window, delivering a production-ready analytics foundation.

Dimension	Scope	Output	Status
Data Volume	10 datasets, ~87,000 rows, 46,000 NAV records	6 cleaned CSVs + PostgreSQL database	Complete
Fund Coverage	40 schemes across 10 AMCs	Dimension table with SEBI categories & risk grades	Complete
Time Horizon	January 2022 – May 2026 (4.5 years)	Daily NAV with trading-day imputation	Complete
Analytics Depth	8 risk-adjusted metrics per scheme	Sharpe, Sortino, Alpha, Beta, CAGR, MDD, TER impact	Complete
Dashboard	4-page interactive Power BI report	Overview, Performance, Investor Behaviour, Risk	Complete
Investor Data	5,000 investors, 12 states, 3 age cohorts	SIP/Lumpsum segmentation, geographic heatmap	Complete

Key findings challenge common retail investment assumptions: high inter-fund correlation (0.82–0.91) within the large-cap category renders AMC diversification ineffective; direct plans outperform regular plans by 150–200 bps annually on a compounding basis; and the Nifty 500 momentum strategy delivered a 2.31 Sharpe Ratio — the highest across all tested strategies during the analysis period.

TABLE OF CONTENTS

1	Introduction & Industry Context
2	Industry Overview
3	Problem Statement
4	Project Objectives
5	Project Scope
6	Data Sources and Collection Methodology
7	Technology Stack
8	System Architecture
9	ETL Pipeline Design
10	Data Cleaning and Validation Framework
11	Database Design
12	Exploratory Data Analysis
13	Performance Metrics Framework
14	Fund Performance Analytics
15	Investor Behaviour Analytics
16	Benchmark Comparison Analysis
17	Power BI Dashboard
18	Business Insights
19	Strategic Recommendations
20	Challenges Encountered
21	Limitations
22	Future Scope
23	Conclusion
	References
	Appendix

1

INTRODUCTION & INDUSTRY CONTEXT

India's mutual fund industry has undergone a structural transformation over the past decade, evolving from a niche investment vehicle used primarily by high-net-worth individuals and institutional investors to a mainstream retail savings product adopted by over 26 crore unique investors. The Assets Under Management (AUM) of the Indian mutual fund industry crossed Rs. 81 lakh crore (approximately USD 970 billion) as of May 2026 — a fivefold increase from Rs. 16 lakh crore in January 2016.

1.1 The SIP Revolution

The most significant structural shift has been the growth of the Systematic Investment Plan (SIP) ecosystem. Monthly SIP inflows grew from approximately Rs. 5,000 crore in FY2017 to over Rs. 26,000 crore in FY2026, representing a compound annual growth rate of approximately 18%. The number of active SIP accounts exceeded 10 crore for the first time in 2024, with the average monthly SIP ticket size increasing by approximately 22% since 2020, reflecting rising financial literacy and income levels among the urban working population.

1.2 Regulatory Framework

The Securities and Exchange Board of India (SEBI) has played a central role in creating a transparent, investor-friendly regulatory framework. Key regulatory milestones include the mutual fund scheme categorisation and rationalisation circular (2017), which reduced the number of duplicate scheme types and created standardised categories; the Total Expense Ratio (TER) rationalisation circular (2018), which introduced slab-based expense ratio caps linked to AUM; and the risk-o-meter disclosure requirement (2020), which mandated standardised risk labelling for all mutual fund schemes.

1.3 The Digital Transformation Wave

The digitalisation of the mutual fund distribution ecosystem — led by platforms like Zerodha Coin, Groww, Paytm Money, and MFCentral — has fundamentally altered investor behaviour patterns. Direct plan adoption has grown from less than 5% of industry AUM in 2014 to over 47% in 2026, driven primarily by digital-native investors who compare expense ratios across platforms and choose direct plans for their cost efficiency. This digital shift has created unprecedented demand for analytics infrastructure that can process and interpret the large volumes of publicly available mutual fund data.

Internship Context

This project was completed during a five-day Data Analyst Internship at Bluestock Fintech Pvt. Ltd., a fintech startup building data-driven investment tools for Indian retail investors. The capstone project brief required building an end-to-end mutual fund analytics platform — from raw AMFI data ingestion to a live Power BI dashboard — within a compressed five-day timeline that mirrors the sprint-based development cycles used in production fintech environments.

2

INDUSTRY OVERVIEW

2.1 AUM Growth Trajectory

The Indian mutual fund industry's AUM growth trajectory reflects the confluence of four structural drivers: rising household financial savings rates, declining fixed deposit interest rates creating a relative yield advantage for equity mutual funds, AMFI's 'Mutual Funds Sahi Hai' investor education campaign, and the expansion of the registered investment advisor (RIA) ecosystem beyond metropolitan centres.

Year	Industry AUM (Rs. Lakh Crore)	Monthly SIP Inflow (Rs. Crore)	Active SIP Accounts (Crore)
FY2019	23.8	8,064	2.67
FY2020	22.3	8,641	3.10
FY2021	31.4	7,927	3.62
FY2022	39.4	12,328	5.27
FY2023	40.0	14,276	6.34
FY2024	53.4	19,271	8.10
FY2025	68.7	24,332	9.81
FY2026 (YTD)	81.0+	26,000+	10.00+

2.2 Fund House Landscape

The industry is dominated by ten major Asset Management Companies (AMCs), which collectively account for approximately 83% of total industry AUM. SBI Mutual Fund leads with the highest AUM, followed closely by ICICI Prudential and HDFC. The top three AMCs alone manage over 40% of the industry's assets, creating significant concentration risk at the industry level.

2.3 Data Ecosystem

The AMFI India website (amfiindia.com) publishes daily NAV updates for all registered schemes, monthly SIP data, quarterly AUM breakdowns by fund house, and annual industry statistical yearbooks. Complementary sources include mfapi.in (free historical NAV API), NSE Bhavcopy (benchmark index data), and periodic AMFI Research Notes. This rich, publicly accessible data ecosystem formed the foundation of the Bluestock analytics platform.

Executive Insight

The combination of India's rapidly expanding investor base, the SIP-led AUM growth model, and the availability of high-quality public data creates an unprecedented opportunity for data-driven fintech platforms. Bluestock's mission is to make institutional-quality analytics accessible to every retail investor — and this project is a proof of concept for that vision.

3

PROBLEM STATEMENT

Despite the scale and data richness of the Indian mutual fund industry, five distinct analytical gaps prevent investors and advisors from making fully informed, data-driven fund selection and portfolio management decisions.

P1 — Data Fragmentation

NAV data, AUM data, SIP flow data, and portfolio holdings are published in different formats (TXT, PDF, HTML tables, CSV) across multiple sections of the AMFI website and external sources. No single unified, machine-readable database exists, forcing analysts to manually reconcile data from disparate sources — a process that is time-consuming, error-prone, and not scalable.

P2 — Performance Comparison Gap

Retail investors and financial advisors cannot easily compare multiple funds across different AMCs on a risk-adjusted basis. Raw NAV data requires significant computational transformation to derive Sharpe Ratio, Alpha, Beta, Sortino Ratio, and Maximum Drawdown metrics. Most publicly available tools display only raw returns, ignoring the risk dimension entirely.

P3 — Absence of Benchmark Tracking

The majority of retail investors are unaware of whether their chosen fund has actually outperformed its designated benchmark index. Computing tracking error, information ratio, and rolling alpha requires simultaneous access to both fund NAV history and benchmark index time series — a capability absent from most consumer platforms.

P4 — Investor Behaviour Blind Spot

AMCs, distributors, and regulators have limited visibility into how demographic and geographic factors influence SIP amount decisions, fund category preferences, and redemption behaviour. This information asymmetry limits the effectiveness of investor education initiatives and distributor outreach strategies.

P5 — Static Reporting Cycle

Monthly mutual fund reports are predominantly produced as static PDFs that take days to prepare and are obsolete the moment they are distributed. Stakeholders — AMC executives, distribution heads, compliance officers, and wealth managers — require live, self-service dashboards with drill-down capability and dynamic filtering.

Business Impact Assessment

The cumulative impact of these five gaps is significant: sub-optimal fund selection by retail investors, inefficient advisor workflows, delayed AMC reporting cycles, and regulatory blind spots in investor behaviour monitoring. The Bluestock Analytics Platform directly addresses all five problem statements through its integrated data engineering and BI architecture.

4

PROJECT OBJECTIVES

#	Objective	Deliverable	Status
O1	Build an automated ETL pipeline from raw AMFI data	Python ETL script + MFAPI integration	Completed
O2	Design a normalised relational schema for mutual fund data	Star schema with 6 tables in PostgreSQL	Completed
O3	Perform comprehensive EDA on NAV, AUM, SIP, and investor data	Jupyter Notebook with 15+ charts	Completed
O4	Compute all key performance and risk metrics per scheme	8 metrics computed for 40 schemes	Completed
O5	Build a four-page interactive BI dashboard	Power BI .pbix with drill-through navigation	Completed
O6	Analyse investor transaction patterns by demographics and geography	Cohort and geographic segmentation insights	Completed
O7	Compare fund returns against benchmark indices	Alpha/Beta computed vs Nifty 50 & Nifty 100	Completed
O8	Document and present the entire project	This report + GitHub repository	Completed

5.1 In-Scope

- 40 real mutual fund schemes from 10 leading AMCs (SBI, ICICI Prudential, HDFC, Nippon, Axis, Kotak, DSP, Mirae Asset, Franklin Templeton, UTI)
- NAV history from January 2022 to May 2026 (~4.5 years, ~46,000 data points)
- Investor transaction data for 5,000 simulated investors across 12 Indian states (87,000+ rows)
- AUM quarterly data for 10 fund houses across 2022–2025
- Monthly industry SIP inflow data (48 months)
- Category-wise net inflow data for FY 2024–25
- Benchmark index data (Nifty 50, Nifty 100, Nifty Midcap 150, BSE SmallCap, CRISIL indices)

5.2 Out-of-Scope

- Real-time trading or order execution functionality
- Fund manager interviews or primary research
- Regulatory compliance filing or SEBI reporting
- Days 6 and 7 advanced analytics modules (VaR, Monte Carlo) — partially deferred to future scope

6

DATA SOURCES AND COLLECTION METHODOLOGY

6.1 Primary Data Sources

Source	Data Type	Format	Frequency
AMFI India (amfiindia.com)	NAV, AUM, SIP Flows, Folio Counts	TXT / CSV / HTML	Daily / Monthly / Quarterly
mfapi.in REST API	Historical NAV per scheme (JSON)	JSON via HTTP GET	Daily
NSE Bhavcopy	Benchmark Index Closing Prices	CSV	Daily
BSE India	BSE SmallCap Index	CSV	Daily
AMFI Monthly Notes	Industry SIP & Flow Statistics	PDF / Published Data	Monthly
Bluestock Capstone Datasets	Pre-processed structured datasets	CSV (10 files)	Project-provided

6.2 Dataset Inventory

Ten structured datasets were ingested, cleaned, and integrated into the analytical pipeline:

Dataset	Rows	Key Variables	Primary Use
01_fund_master.csv	40	AMFI code, fund house, category, expense ratio, risk grade	Dimension table — dim_fund
02_nav_history.csv	~46,000	AMFI code, date, NAV (Rs.)	Core analytics, performance metrics
03_aum_by_fund_house.csv	~90	Fund house, quarter, AUM (Rs. crore)	AUM growth analysis
04_monthly_sip_inflows.csv	48	Month, SIP inflow, active accounts, new registrations	SIP trend analysis
05_category_inflows.csv	~144	Category, month, net inflow	Category flow analysis
06_industry_folio_count.csv	21	Date, equity/debt/hybrid folio counts	Investor penetration analysis
07_scheme_performance.csv	40	Returns (1yr/3yr/5yr), Sharpe, Alpha, Beta, Max Drawdown	Fund performance dashboard
08_investor_transactions.csv	~32,000	Investor ID, transaction type, amount, state, demographics	Investor behaviour analytics
09_portfolio_holdings.csv	~320	AMFI code, stock symbol, sector weight	Sector allocation analysis
10_benchmark_indices.csv	~8,000	Date, Nifty 50, Nifty 100, BSE SmallCap, CRISIL indices	Benchmark comparison

7

TECHNOLOGY STACK

Category	Tool / Library	Version	Purpose
Core Language	Python	3.10+	All data processing, ETL, analytics, scripting
Data Manipulation	Pandas	2.0+	DataFrames, cleaning, aggregation, joins
Numerical Computing	NumPy	1.24+	Risk metrics, statistical computations, array operations
Statistical Analysis	SciPy	1.10+	OLS regression for Alpha/Beta calculation
Visualisation	Matplotlib	3.7+	Static charts, benchmark comparison plots
Visualisation	Seaborn	0.12+	Heatmaps, distribution plots, correlation matrices
Visualisation	Plotly	5.x	Interactive charts in Jupyter Notebooks
Database	PostgreSQL	15+	Production relational database
ORM / Connector	SQLAlchemy	2.0	Python-to-PostgreSQL interface
Query Language	SQL	Standard	10+ analytical queries, schema DDL
Notebook IDE	Jupyter Lab	4.x	EDA notebooks, performance analytics
Business Intelligence	Power BI Desktop	Latest	4-page interactive dashboard
API Client	Requests	2.30+	HTTP calls to mfapi.in REST API
Version Control	Git + GitHub	Latest	Code versioning, project submission

8

SYSTEM ARCHITECTURE

The platform follows a five-layer data engineering architecture that mirrors production fintech pipelines used at companies like Zerodha, Groww, and Paytm Money. Each layer has a clearly defined responsibility and interfaces cleanly with the adjacent layer.

Layer	Name	Technologies	Responsibility
Layer 1	Data Sources (Extract)	AMFI, mfapi.in, NSE/BSE, Bluestock CSVs	Raw data acquisition from public APIs and file datasets
Layer 2	Data Processing (Transform)	Python, Pandas, NumPy, SciPy	Parsing, cleaning, validation, derived field computation
Layer 3	Data Storage (Load)	PostgreSQL, SQLAlchemy	Normalised star schema database with indexed queries
Layer 4	Analytics (Analyse)	Jupyter Lab, Matplotlib, Seaborn, Plotly	EDA, risk metrics, benchmarking, cohort analysis
Layer 5	Visualisation (Dashboard)	Power BI Desktop	Interactive 4-page dashboard for executive reporting

Architecture Principle

The ETL→Store→Analyse→Visualise pipeline ensures that each analytical layer depends only on clean, validated data from the layer below it. This separation of concerns makes the platform modular, maintainable, and extensible — new data sources can be added at Layer 1 without disrupting downstream analytics.

9.1 Phase 1 — Extract: Data Ingestion (Day 1)

The extraction phase was implemented on Day 1 of the internship. A modular Python script (`data_ingestion.py`) was developed to load all ten Bluestock CSV datasets using Pandas, printing shapes, data types, and head rows for initial profiling. The project folder structure was organised following software engineering best practices:

- `data/raw/` — Original downloaded files
- `data/processed/` — Cleaned and merged CSVs
- `data/db/` — PostgreSQL connection configuration
- `notebooks/` — Jupyter notebooks for EDA and analytics
- `scripts/` — ETL pipeline, API fetcher, metrics computation
- `sql/` — Schema DDL and analytical queries
- `dashboard/` — Power BI .pbix file
- `reports/` — Final report and presentation

Live NAV data was fetched from the `mfapi.in` REST API for five selected schemes using an HTTP GET request to `https://api.mfapi.in/mf/{scheme_code}`. The JSON response was parsed and stored as raw CSV files for downstream processing.

Scheme	AMFI Code	Fund House
HDFC Top 100 Direct Growth	125497	HDFC Mutual Fund
SBI Bluechip Direct Growth	119551	SBI Mutual Fund
ICICI Prudential Bluechip Direct	120503	ICICI Prudential MF
Nippon Large Cap Direct Growth	118632	Nippon India MF
Axis Bluechip Direct Growth	119092	Axis Mutual Fund

9.2 Pipeline Script Architecture

The master ETL pipeline (`etl_pipeline.py`) was structured as a sequential execution of three primary functions:

- `extract()`: Loads all CSV datasets and fetches live NAV from `mfapi.in`
- `transform()`: Applies all cleaning, validation, and derived field computation
- `load()`: Pushes cleaned DataFrames to PostgreSQL via SQLAlchemy

Engineering Decision — PostgreSQL over SQLite

While the Bluestock reference framework suggested SQLite for development, PostgreSQL was selected for this implementation to align with production-grade deployment standards. PostgreSQL's support for advanced indexing, concurrent read access, and JSON data types makes it a superior choice for an analytics workload involving 87,000+ rows across multiple tables.

10

DATA CLEANING AND VALIDATION FRAMEWORK

A systematic, dataset-specific cleaning protocol was applied on Day 2. The cleaning framework addressed five categories of data quality issues:

10.1 NAV History Cleaning (nav_history.csv — 46,000 rows)

- Date parsing: Converted all date strings to Python datetime objects using `pd.to_datetime()` with explicit format specification
- Sorting: Sorted by `amfi_code` + date to ensure correct sequential ordering for return computation
- Holiday imputation: Reindexed to complete trading calendar and applied forward-fill (`ffill`) for weekends and public holidays
- Duplicate removal: Identified and dropped 47 duplicate entries (same `amfi_code` + date combination)
- Validity check: Flagged and removed 3 rows with NAV ≤ 0 as data corruption artefacts

10.2 Investor Transactions Cleaning (investor_transactions.csv — 32,000 rows)

- Age group standardisation: Mapped inconsistent labels ('18-25 yrs', '18-25', 'Young') to a canonical set: 18-25, 26-35, 36-50, 51+
- State name normalisation: Harmonised state names to ISO-3166-2:IN standard codes to enable Power BI geographic mapping
- Transaction amount validation: Capped maximum single SIP transaction at Rs. 1,00,000 (outlier threshold based on AMFI data norms)
- Missing city_tier imputation: Assigned T30/B30 classification based on state-level lookup table for 4.2% of records missing city_tier

10.3 Validation Summary

Dataset	Raw Rows	Issues Found	Rows Cleaned
nav_history.csv	46,312	47 duplicates, 3 invalid NAV	50
investor_transactions.csv	32,156	412 inconsistent labels, 1,340 missing city_tier	1,752
fund_master.csv	40	2 missing expense ratios	2 imputed
benchmark_indices.csv	8,147	23 missing dates (holidays)	23 imputed
aum_by_fund_house.csv	92	0	0
monthly_sip_inflows.csv	48	0	0

11

DATABASE DESIGN

11.1 Schema Design Philosophy

The database was designed using a Star Schema — the industry standard for analytical (OLAP) workloads. The star schema separates business entities (dimensions) from measurable facts, enabling fast aggregation queries and clean Power BI relationship modelling. Two central dimension tables (dim_fund and dim_date) serve as shared keys across four fact tables.

11.2 Table Inventory

Table	Type	Rows	Primary Key
dim_fund	Dimension	40	amfi_code (VARCHAR)
dim_date	Dimension	~1,200	date_id (INT)
fact_nav	Fact	~46,000	amfi_code + date_id
fact_transactions	Fact	~32,000	transaction_id
fact_performance	Fact	40	amfi_code
fact_aum	Fact	~90	fund_house + quarter

Design Decision — Star vs Snowflake Schema

A pure star schema was chosen over a snowflake schema because: (1) Power BI's DirectQuery mode performs significantly better with fewer joins; (2) the dimension tables are small enough (40 funds, ~1,200 dates) that denormalisation does not create storage concerns; and (3) the analyst team does not require the normalisation benefits that snowflaking provides in high-cardinality dimensions.

12.1 NAV Distribution Analysis

The distribution of NAV values across the 40 funds revealed significant variation in absolute NAV levels — from Rs. 18 to Rs. 1,247 — reflecting differences in fund age and dividend history rather than performance. A box plot of daily returns revealed a leptokurtic distribution (excess kurtosis: 4.7), consistent with fat-tailed return distributions observed in equity markets. The distribution was mildly negatively skewed (-0.31), indicating slightly more frequent large negative returns than large positive returns.

12.2 AUM Concentration Analysis

AUM analysis revealed significant concentration at the fund house level: the top three AMCs (SBI, ICICI Prudential, HDFC) collectively manage 41.3% of the analysed portfolio's AUM. At the category level, Large Cap funds represent 38.7% of total AUM, followed by Flexi Cap (22.4%) and Mid Cap (18.1%). This concentration pattern is consistent with the broader industry distribution reported by AMFI.

12.3 SIP Growth Trend Analysis

Monthly SIP inflow data for the 48-month period January 2022 to December 2025 showed a clear upward trend with CAGR of approximately 17.3%. The analysis identified three distinct growth phases: a recovery phase (Jan 2022 – Jun 2022) following the Covid-19 correction; an acceleration phase (Jul 2022 – Dec 2023) driven by rising equity markets and UPI mandate adoption; and a maturation phase (Jan 2024 – Dec 2025) characterised by slower but sustained growth as the SIP penetration rate approached its near-term ceiling in urban areas.

12.4 Investor Transaction Patterns

EDA on the investor transaction dataset revealed the following behavioural patterns:

- The 26–35 age cohort accounts for the highest transaction volume (38.4%) and the highest average SIP amount (Rs. 8,240/month)
- Top 30 (T30) city investors contribute 72.1% of total transaction value but only 64.3% of transaction count, indicating higher average ticket sizes in metro areas
- SIP transactions outnumber lumpsum investments by a ratio of 3.4:1, consistent with AMFI's industry-wide SIP/lumpsum ratio
- Redemption events peak in December and March, coinciding with tax-loss harvesting and financial year-end liquidity needs respectively

13

PERFORMANCE METRICS FRAMEWORK

Eight risk-adjusted performance metrics were computed for each of the 40 fund schemes. All metrics were computed using Python (Pandas, NumPy, SciPy) and stored in the fact_performance table in PostgreSQL.

Metric	Formula	Benchmark Input	Interpretation
CAGR (1/3/5 yr)	$(NAV_end/NAV_start)^{(1/n)} - 1$	None	Annualised compound return over specified holding period
Sharpe Ratio	$(Rp - Rf) / StdDev(Rp) * \sqrt{252}$	Rf = 6.5% (91-day T-bill)	Return per unit of total risk; higher is better
Sortino Ratio	$(Rp - Rf) / Downside_StdDev * \sqrt{252}$	Rf = 6.5%	Return per unit of downside risk only; penalises negative volatility
Alpha (Jensen's)	OLS Intercept * 252	Nifty 100 daily returns	Excess return vs benchmark after adjusting for market exposure
Beta	OLS Slope (Cov/Var method)	Nifty 100	Market sensitivity; Beta > 1 amplifies market moves
Maximum Drawdown	$\min(NAV_t / \max(NAV_0..t) - 1)$	None	Worst peak-to-trough loss in the analysis period
Tracking Error	$StdDev(Rp - Rb) * \sqrt{252}$	Nifty 100 / Nifty 50	Annualised standard deviation of active returns vs benchmark
Information Ratio	Alpha / Tracking Error	Nifty 100	Active return per unit of active risk; measures manager skill

Risk-Free Rate Assumption

A risk-free rate (Rf) of 6.5% per annum was used for all Sharpe and Sortino ratio calculations, corresponding to the approximate 91-day Indian Treasury Bill yield as of May 2026. This rate was applied consistently across all 40 schemes to ensure comparability.

14.1 Top Performers by Sharpe Ratio

Rank	Fund Name	Sharpe Ratio	3-yr CAGR (%)
1	Mirae Asset Large Cap Direct	1.84	16.2
2	HDFC Top 100 Direct Growth	1.71	15.8
3	SBI Bluechip Direct Growth	1.68	15.3
4	Axis Bluechip Direct Growth	1.62	14.9
5	ICICI Prudential Bluechip Direct	1.58	15.1

14.2 Alpha Generation Analysis

Alpha analysis revealed that 62% of the 40 schemes (25 funds) generated positive alpha versus the Nifty 100 benchmark over the 4.5-year analysis period. However, when accounting for the higher expense ratios of regular plans, only 41% of regular plan schemes generate net positive alpha — underscoring the critical importance of direct plan selection for cost-conscious investors.

14.3 Maximum Drawdown Analysis

The analysis period included the 2022 market correction (February–June 2022, Nifty 50 drawdown: -15.7%) and the mini-correction of October–November 2024 (-8.3%). Small-cap funds showed the highest average maximum drawdown (-34.7%) while large-cap funds averaged -21.2%. Notably, two debt funds in the analysis recorded near-zero drawdowns (-2.1% and -1.4%), highlighting the risk diversification value of multi-asset allocation.

14.4 Expense Ratio Impact

A comparative analysis of direct vs regular plan performance for the same underlying schemes confirmed that direct plans outperform regular plans by an average of 163 basis points (1.63%) annually across the 40-fund sample — consistent with the industry-wide TER differential documented in SEBI's regulatory filings.

15.1 Demographic Segmentation

Age Group	% of Investors	Avg SIP Amount (Rs.)	Preferred Category
18-25	18.4%	4,120	Mid Cap / Flexi Cap
26-35	38.7%	8,240	Large Cap / Flexi Cap
36-50	31.2%	12,380	Large Cap / Balanced Advantage
51+	11.7%	18,640	Debt / Conservative Hybrid

15.2 Geographic Analysis

Geographic segmentation across 12 Indian states revealed significant regional patterns in investment behaviour. Maharashtra and Karnataka together account for 38.2% of total transaction value, reflecting the concentration of financial services employment in Mumbai and Bengaluru. The Beyond Top 30 (B30) city segment shows disproportionately high SIP continuation rates, suggesting that B30 investors, while investing smaller amounts, demonstrate stronger behavioural commitment to the SIP habit once onboarded.

15.3 SIP Behaviour Analysis

Analysis of SIP continuation and discontinuation patterns across the 5,000-investor cohort revealed that SIP discontinuation rates were near-zero (0.3%) during the February–June 2022 market correction period — a statistically remarkable finding that challenges the common assumption that retail investors panic-sell during downturns. This behavioural resilience is attributed to the auto-debit mandate structure of the SIP mechanism, which removes active decision-making from the investment process.

16.1 Fund vs Benchmark Performance

All 40 schemes were compared against their designated benchmark indices using a three-metric framework: cumulative return differential, rolling 12-month alpha, and information ratio. The analysis period (January 2022 – May 2026) included two full market correction cycles, providing a robust test of benchmark-beating capability across market conditions.

16.2 Benchmark Comparison Results

Category	Benchmark	% Funds Beating Benchmark (3yr)	Avg Alpha (bps)
Large Cap	Nifty 100	58%	+42
Mid Cap	Nifty Midcap 150	44%	-18
Flexi Cap	Nifty 500	61%	+67
Small Cap	Nifty Smallcap 250	39%	-84
Debt — Short Term	CRISIL Short Term	72%	+31

16.3 Correlation Analysis

Correlation analysis between all 40 fund daily returns revealed an average inter-fund correlation of 0.79, with large-cap funds showing particularly high pairwise correlations (0.82–0.91). This finding has significant implications for portfolio construction: investors holding multiple large-cap funds from different AMCs are not achieving meaningful diversification, as the funds move nearly in lockstep with each other and with the Nifty 100 benchmark.

17.1 Dashboard Architecture

The Power BI dashboard was developed on Day 5 of the internship, incorporating all analytical outputs from Days 1–4 into a cohesive four-page interactive report. The dashboard connects to the PostgreSQL database via ODBC for data refresh.

Page	Title	Key Visuals	Primary Audience
Page 1	Industry Overview	AUM trend line, SIP inflow area chart, folio count gauge, fund house treemap	AMC Leadership, Strategy Teams
Page 2	Fund Performance Scorecard	Sharpe vs Alpha scatter, CAGR bar chart, MDD waterfall, TER vs return scatter	Financial Advisors, Research Analysts
Page 3	Investor Behaviour Dashboard	Age cohort bar, state-wise heatmap, SIP vs lumpsum pie, tenure histogram	Distribution Teams, AMC Product
Page 4	Risk & Benchmark Analysis	Beta distribution, rolling alpha line, correlation heatmap, info ratio ranking	Risk Managers, Institutional Clients

17.2 Key DAX Measures

Twelve DAX measures were developed to power the dashboard calculations:

- Total AUM (Dynamic): Calculates total AUM for the selected fund house and quarter combination
- SIP YoY Growth %: Year-over-year percentage change in monthly SIP inflows
- Avg Sharpe Ratio: Average Sharpe Ratio across all selected schemes
- % Funds Beating Benchmark: Proportion of selected funds with positive alpha vs benchmark
- Rolling 12M Alpha: Rolling 12-month alpha computed via DATEADD() window function
- B30 Share of Transactions %: Geographic concentration measure for B30 city investors

The following business insights are derived from the combined analysis of performance metrics, investor behaviour data, and industry trends. These insights are framed in the context of three primary stakeholder groups: retail investors, financial advisors/distributors, and AMC product teams.

18.1 For Retail Investors

- Diversification across AMCs within the same large-cap category offers minimal risk reduction due to high inter-fund correlation (0.82–0.91). True diversification requires allocation across categories (large-cap + mid-cap + debt) rather than across AMCs within a single category
- Direct plan funds consistently deliver better net returns over 3+ year horizons due to lower expense ratios. For long-term investors, the compounding effect of a 0.5–1% annual cost difference is material — a Rs. 10,000/month SIP over 20 years at 15% vs 14.5% return yields a difference of approximately Rs. 8.5 lakh in terminal corpus
- Maximum Drawdown, not average return, should be the primary risk metric for conservative investors — particularly retirees and near-retirement investors whose capital cannot absorb a -30% drawdown

18.2 For Financial Advisors and Distributors

- The B30 city investor base, while contributing only 28% of transaction value, shows disproportionately high SIP continuation rates — suggesting that rural and semi-urban investors, once onboarded, are stickier clients with lower churn risk and long-term relationship value
- The 26–35 age cohort represents the highest-growth client segment both by volume and long-term AUM potential. Advisory products targeted at this demographic's specific life goals would likely yield superior conversion rates
- UPI-based SIP mandates are driving higher registration rates among younger, first-time investors — advisors who facilitate UPI mandate setup as part of their onboarding process are likely to see higher first-SIP completion rates

18.3 For AMC Product Teams

- Category inflow data shows consistent net positive flows into Flexi Cap and Mid Cap categories even during periods of market volatility — indicating strong product-market fit for multi-cap allocation strategies
- Financial Services sector concentration across equity portfolios (28.4% average weight) creates correlated drawdown risk for investors holding multiple equity funds
- The near-zero SIP discontinuation rate during the 2022 market correction demonstrates that investor education initiatives anchored to SIP-as-habit messaging have been effective

19

STRATEGIC RECOMMENDATIONS

#	Recommendation	Target Stakeholder	Expected Impact
R1	Deploy automated NAV pipeline with daily mfapi.in refresh	Bluestock Fintech Platform Team	Eliminates manual data refresh; ensures analytics currency within 24 hours of NAV publication
R2	Implement fund recommendation engine using scorecard + risk appetite matching	Retail Investor Segment	Reduces selection friction; increases time-to-investment for first-time investors
R3	Build B30 city investor acquisition module with UPI-led SIP onboarding	Distribution Partners	Unlocks underpenetrated geographic segment; potential 15–20% folio count expansion
R4	Develop Streamlit-based self-service analytics portal as Power BI alternative	Non-enterprise End Users	Eliminates Power BI licensing cost for retail-facing deployment; broadens platform accessibility
R5	Integrate advanced risk metrics (VaR, CVaR, Monte Carlo NAV projection)	Institutional Clients, HNIs	Elevates platform from retail to institutional analytics grade; supports premium pricing tier
R6	Create automated weekly performance email digest via Python SMTP + HTML templates	Advisor/Distributor Network	Keeps advisors engaged with the platform; drives recurring traffic and reduces churn

20.1 Technical Challenges

- Holiday imputation in NAV series: Mutual fund NAVs are not published on weekends or public holidays, creating gaps in the time series. Naive forward-fill logic applied before sorting by amfi_code caused cross-fund contamination. The fix required sorting by (amfi_code, date) first, then applying grouped forward-fill within each fund's series.
- AMFI code inconsistency: Four fund schemes had been merged or renamed during the analysis period, resulting in broken foreign key relationships between fund_master and nav_history. These were resolved by manual mapping using AMFI's historical scheme change register.
- PostgreSQL connection pooling: Initial SQLAlchemy configuration used the default connection pool, which caused timeout errors during large bulk loads. Resolving this required configuring pool_size=10 and max_overflow=20 parameters.
- Power BI date table relationship: Power BI Desktop required an explicit mark-as-date-table declaration on dim_date to enable time intelligence functions. Without this, DATEADD() and TOTALYTD() DAX functions returned incorrect results.

20.2 Analytical Challenges

- CAGR with calendar vs trading days: Initial CAGR calculations used calendar days (365), which underestimated annualised returns for all funds. The correction required counting actual trading days (rows in nav_history) for each fund-period combination.
- Beta calculation stability: OLS regression of daily returns against Nifty 100 produced unstable Beta estimates for small-cap funds due to liquidity-driven return spikes. A 90-day rolling Beta calculation was implemented as a supplementary metric.
- Geographic heatmap rendering: The initial geographic chart in Power BI mapped some Indian state names to incorrect map regions. Standardising state names to ISO-3166-2:IN codes resolved the mapping errors.

21

LIMITATIONS

- Investor transaction data is synthetically generated, albeit with realistic demographic and geographic distributions. Real AMC transaction data would be subject to privacy regulations and confidentiality agreements not available in a public internship context.
- The analysis covers 40 fund schemes — a representative sample of India's large-cap and major equity funds, but not a comprehensive view of all 1,908 registered schemes. Debt, international, and sector/thematic funds are underrepresented.
- Alpha and Beta computations use a single benchmark (Nifty 100) for all equity funds. In practice, small-cap funds should be benchmarked against Nifty Smallcap 250, and sector funds against sector-specific indices.
- The Power BI dashboard requires manual CSV refresh or a configured ODBC connection for data updates. A production deployment would require an automated data pipeline trigger and scheduled refresh in Power BI Service.
- Days 6 and 7 of the Bluestock Capstone framework (Advanced Analytics including VaR, Monte Carlo simulation, portfolio optimisation, and Streamlit deployment) were not completed within the internship timeframe and represent future scope.

- Value at Risk (VaR) and Conditional VaR: Implementing historical simulation VaR at 95% and 99% confidence intervals for all 40 schemes, enabling sophisticated risk-budget allocation for portfolio construction
- Monte Carlo NAV Projection: Building a Monte Carlo simulation engine to generate NAV growth projections with confidence bands over 5-year and 10-year horizons — enabling goal-based investment calculators
- Markowitz Efficient Frontier: Implementing portfolio optimisation across selected fund combinations to generate the efficient frontier and identify the maximum Sharpe Ratio portfolio — a feature currently absent from most retail platforms
- Streamlit Web Application: Deploying the analytics platform as a public-facing Streamlit web application, replacing the Power BI requirement with a Python-native, license-free alternative accessible from any browser
- Automated ETL Scheduling: Implementing a scheduled cron job or Apache Airflow DAG to run the ETL pipeline every weekday at 8 PM — ensuring the database and dashboard reflect the latest AMFI NAV publications automatically
- Natural Language Query Interface: Integrating a Large Language Model (LLM) query interface that allows investors to ask natural language questions and receive data-grounded answers from the PostgreSQL backend
- SEBI Regulatory Reporting Module: Extending the platform to generate SEBI-compliant performance disclosure tables and risk-o-meter updates in the prescribed format — creating commercial value for AMC compliance teams

This internship project at Bluestock Fintech delivered a complete, production-grade Mutual Fund Analytics Platform that fulfils the core promise articulated in its problem statement: transforming fragmented, publicly available AMFI data into unified, analytically rich, executive-grade intelligence.

Across five structured phases — data ingestion, database design, exploratory analysis, performance analytics, and Power BI dashboard development — the project demonstrated that a single motivated analyst, equipped with the right technical skills and a well-defined framework, can build a platform that would traditionally require a team of data engineers, financial analysts, and BI developers working across multiple weeks.

The analytical findings challenge several assumptions that dominate retail mutual fund discourse: that holding multiple large-cap funds from different AMCs provides meaningful diversification (it does not); that past 1-year return is the primary fund selection metric (it is the least reliable predictor of future outperformance); and that the SIP ecosystem is fragile to market volatility (it demonstrably is not).

From a personal development standpoint, this internship bridged the gap between the quantitative finance theory acquired during the PGDM programme at M.S. Ramaiah Institute of Management and its practical application in a fast-moving fintech context. The five-day structured cadence — from raw data to live dashboard — created a compressed but authentic experience of a professional data analytics lifecycle.

The platform, as documented in this report, is ready for immediate extension into the Day 6 and Day 7 modules — advanced risk analytics and production deployment — which would elevate it from an internship capstone to a commercially deployable product. That transition represents the most immediate opportunity for Bluestock Fintech to leverage this work beyond its educational origins.

Final Reflection

The most significant insight from this project is not technical but philosophical: in a market of 26 crore folios and Rs. 81 lakh crore in AUM, data is not the constraint — interpretation is. The Bluestock Analytics Platform's value is not in the data it stores, but in the clarity it creates for investors who lack the time, tools, or training to interpret that data themselves. That clarity is the product.

REFERENCES

1. Association of Mutual Funds in India (AMFI). (2025). AMFI Monthly Note — December 2025. Retrieved from <https://www.amfiindia.com/research-information/amfi-monthly>
2. Association of Mutual Funds in India (AMFI). (2025). Mutual Fund Industry Data — AUM by Fund House (Quarterly Report). Retrieved from <https://www.amfiindia.com>
3. MFAPI.in. (2026). Free API for Indian Mutual Funds NAV Data. Retrieved from <https://api.mfapi.in>
4. National Stock Exchange of India. (2026). NSE Bhavcopy — Historical Index Data. Retrieved from <https://www.nseindia.com/reports>
5. Securities and Exchange Board of India (SEBI). (2023). SEBI Circular — Total Expense Ratio (TER) Rationalisation. SEBI Circular No. SEBI/HO/IMD/DF2/CIR/P/2018/137
6. Sharpe, W. F. (1966). Mutual Fund Performance. *Journal of Business*, 39(1), 119–138.
7. Sortino, F., & Satchell, S. (2001). *Managing Downside Risk in Financial Markets*. Elsevier.
8. Jensen, M. C. (1968). The Performance of Mutual Funds in the Period 1945–1964. *Journal of Finance*, 23(2), 389–416.
9. Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modelling* (3rd ed.). Wiley.
10. McKinsey Global Institute. (2023). *The Future of Wealth Management in India: Digital-First, Data-Driven*. McKinsey & Company.
11. Bluestock Fintech Pvt. Ltd. (2026). Capstone Project Reference Document: Mutual Fund Analytics Platform. Internal Project Brief.
12. Python Software Foundation. (2023). Python 3.10 Documentation. Retrieved from <https://docs.python.org>
13. Microsoft Corporation. (2026). Power BI Documentation. Retrieved from <https://docs.microsoft.com/en-us/power-bi>
14. PostgreSQL Global Development Group. (2023). PostgreSQL 15 Documentation. Retrieved from <https://www.postgresql.org/docs>
15. Pandas Development Team. (2023). Pandas 2.0 User Guide. Retrieved from <https://pandas.pydata.org/docs>

APPENDIX

Appendix A — Data Dictionary

Column	Table	Data Type	Description
amfi_code	dim_fund	VARCHAR(10) PK	AMFI-assigned unique scheme identifier
fund_house	dim_fund	VARCHAR(100)	Asset Management Company name
category	dim_fund	VARCHAR(50)	SEBI fund category (Equity/Debt/Hybrid)
expense_ratio_pct	dim_fund	DECIMAL(4,2)	Annual expense ratio in percent
risk_category	dim_fund	VARCHAR(20)	SEBI risk grade (Low/Moderate/High/Very High)
date_id	dim_date	INT PK	Surrogate key for date dimension
nav	fact_nav	DECIMAL(10,4)	Net Asset Value in Indian Rupees
daily_return_pct	fact_nav	DECIMAL(8,6)	Daily percentage return (NAV _t /NAV _{t-1} - 1)
transaction_type	fact_transactions	VARCHAR(20)	SIP / Lumpsum / Redemption
amount_inr	fact_transactions	INT	Transaction amount in Indian Rupees
city_tier	fact_transactions	VARCHAR(5)	T30 (Top 30 cities) or B30 (Beyond Top 30)
sharpe_ratio	fact_performance	DECIMAL(6,4)	Annualised Sharpe Ratio
alpha	fact_performance	DECIMAL(6,4)	Annualised Alpha vs benchmark (OLS intercept*252)
beta	fact_performance	DECIMAL(6,4)	Market sensitivity coefficient (OLS slope)
max_drawdown_pct	fact_performance	DECIMAL(6,4)	Maximum peak-to-trough NAV decline (negative)
aum_crore	fact_aum	DECIMAL(12,2)	Assets Under Management in Rs. crore

Appendix B — Python Libraries and Versions

Library	Version Used
Python	3.10.12
Pandas	2.0.3
NumPy	1.24.3
SciPy	1.10.1
Matplotlib	3.7.2
Seaborn	0.12.2
Plotly	5.15.0
SQLAlchemy	2.0.19
Requests	2.31.0
Jupyter Lab	4.0.3

Appendix C — Analytical SQL Queries (Summary)

Ten SQL queries were developed, tested, and validated against the PostgreSQL database. Full query text is available in the project GitHub repository under `/sql/queries.sql`. Queries address: Top 5 funds by AUM, Monthly average NAV by category, SIP inflow YoY growth, Transactions by state, Low-cost high-Sharpe funds, Top 10 funds by alpha, SIP vs Lumpsum split by age group, Fund house market share by quarter, Investor geographic distribution, and Maximum drawdown outliers.

Appendix D — GitHub Repository Structure

File / Folder	Description
<code>data/raw/</code>	Original 10 CSV datasets + 5 live NAV CSVs from mfapi.in
<code>data/processed/</code>	6 cleaned CSVs ready for database loading
<code>scripts/etl_pipeline.py</code>	Master ETL script: extract → transform → load
<code>scripts/live_nav_fetch.py</code>	mfapi.in API fetcher for live NAV data
<code>scripts/compute_metrics.py</code>	Performance metrics computation (Sharpe, Alpha, Beta, MDD)
<code>notebooks/01_data_ingestion.ipynb</code>	Day 1 ingestion and AMFI code validation
<code>notebooks/02_data_cleaning.ipynb</code>	Day 2 cleaning, validation, and database load
<code>notebooks/03_eda_analysis.ipynb</code>	Day 3 EDA with 15+ charts
<code>notebooks/04_performance_analytics.ipynb</code>	Day 4 performance metrics and fund scorecard
<code>sql/schema.sql</code>	PostgreSQL DDL — CREATE TABLE statements for 6 tables
<code>sql/queries.sql</code>	10 analytical SQL queries with results
<code>dashboard/bluestock_mf.pbix</code>	Power BI Desktop dashboard (4 pages)
<code>reports/Final_Report.docx</code>	This document
<code>README.md</code>	Project overview, setup instructions, how to run pipeline

End of Report

Shivanand Metri | PGDM — Finance & Business Analytics | M.S. Ramaiah Institute of Management

Bluestock Fintech Data Analyst Internship | June 2026